

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

Section 1 - Identification

Product Name

Sulfuric acid

Product Code

CDAIMM207-0192, CDAIMM205-0192, CDAIMM212-0192, CDAIMM205-0480

Address

ThermoFisher Scientific Australia Pty Ltd
 5 Caribbean Drive, Scoresby
 VICTORIA 3179, Australia

Emergency Tel.

**CHEMTREC®
 03 9757 4559 or +613 9757 4559**

Telephone / Fax Numbers

Tel: 1300 735 292
 Fax: 1800 067 639

E-mail address

ANZinfo@thermofisher.com

Recommended Use

In vitro diagnostic.

Uses advised against

This product contains one or more substance(s) on the Illicit Drug Precursors/Reagents list. Verify requirements related to using, handling and storing these substances. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product contains one or more substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

Section 2 - Hazard(s) Identification

Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

Physical hazards

No hazards identified

Health hazards

Skin Corrosion/Irritation
 Serious Eye Damage/Eye Irritation
 Reproductive Toxicity
 Specific target organ toxicity - (single exposure)

Category 2
 Category 1
 Category 1B
 Category 3

Environmental hazards

No hazards identified

Label Elements



Health Hazard



Corrosion

Signal Word**Danger****Hazard Statements**

H315 - Causes skin irritation

H318 - Causes serious eye damage

H335 - May cause respiratory irritation

H360 - May damage fertility or the unborn child

Precautionary Statements

P403 - Store in a well-ventilated place

P501 - Dispose of contents/ container to an approved waste disposal plant

Other information

This product does not contain any known or suspected endocrine disruptors

Section 3 - Composition and Information on Ingredients

Component	CAS No	Weight %
Water	7732-18-5	87
Glycerin	56-81-5	5
D-Mannitol	69-65-8	2
Tris (hydroxymethyl) aminomethane	77-86-1	1
Albumins, blood serum	9048-46-8	1
Sodium chloride	7647-14-5	0.8
Polyoxyethylene(20)sorbitan monolaurate	9005-64-5	0.5
Citric acid	77-92-9	0.2
Phosphoric acid, disodium salt, dodecahydrate	10039-32-4	0.12
[1,1'-Biphenyl]-4,4'-diamine, 3,3',5,5'-tetramethyl-	54827-17-7	0.1
Sulfuric acid	7664-93-9	STOP Solution
Sodium azide	26628-22-8	0.1
Ethylenediamine tetraacetic acid (EDTA)	60-00-4	0.1
Urea peroxide	124-43-6	0.05
ProClin 300	NA	0.04
Potassium chloride	7447-40-7	0.02
Dihydrogen potassium phosphate	7778-77-0	0.02

Section 4 - First Aid Measures

Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

Ingestion

Clean mouth with water and drink afterwards plenty of water.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
General Advice	If symptoms persist, call a physician.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.
First Aid Facilities	Eyewash, safety shower and washroom.
Most important symptoms and effects	None reasonably foreseeable. Causes severe eye damage.
Notes to Physician	Treat symptomatically.

Section 5 - Fire Fighting Measures

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

Section 6 - Accidental Release Measures

Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

Environmental Precautions

Should not be released into the environment.

Methods for Containment and Clean Up

Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

Section 7 - Handling and Storage

Precautions for Safe Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation.

Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed. Keep at temperatures between 2°C and 8 °C.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

Section 8 - Exposure Controls and Personal Protection

Exposure limits

AUS - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

ACGIH - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

UK - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

DE - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

NZ - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

Component	Australia	New Zealand WEL	ACGIH TLV	The United Kingdom	Germany
Glycerin	TWA: 10 mg/m ³	TWA: 10 mg/m ³		TWA: 10 mg/m ³ 8 hr (mist only)	TWA: 200 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 200 mg/m ³ (8 Stunden). MAK Höhepunkt: 400 mg/m ³
Citric acid					TWA: 2 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 2 mg/m ³ (8 Stunden). MAK Höhepunkt: 4 mg/m ³
Sulfuric acid	STEL: 3 mg/m ³ TWA: 1 mg/m ³	TWA: 0.1 mg/m ³	TWA: 0.2 mg/m ³	STEL: 0.15 mg/m ³ 15 min TWA: 0.05 mg/m ³ 8 hr	TWA: 0.1 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 0.1 mg/m ³ (8 Stunden). MAK Höhepunkt: 0.1 mg/m ³
Sodium azide	CL 0.11 ppm (0.3 mg/m ³)	Ceiling: 0.11 ppm Ceiling: 0.29 mg/m ³	Ceiling: 0.29 mg/m ³ Ceiling: 0.11 ppm	Skin TWA 0.1 mg/m ³ STEL 0.3 mg/m ³	MAK 0.2 mg/m ³ (inhalable)

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Exposure Controls**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment**Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	AUS/NZ Standard	Glove comments
Disposable gloves	See manufacturers recommendations	-	AS/NZS 2161	(minimum requirement)

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Skin and body protection

Long sleeved clothing

Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent)
When RPE is used a face piece Fit Test should be conducted

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls

No information available.

Section 9 - Physical and Chemical Properties

Information on basic physical and chemical properties

Appearance	Varies	
Physical State	Liquid	
Odor	No information available	
Odor Threshold	No data available	
pH	Not applicable 4.1	
Melting Point/Range	-40 °C / -40 °F	
Softening Point	No data available	
Boiling Point/Range	Not applicable 189 °C / 372.2 °F	
Flash Point	Not applicable 118 °C / 244.4 °F	Method - No information available
Evaporation Rate	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Vapor Pressure	No data available	
Vapor Density	No data available	(Air = 1.0)
Specific Gravity / Density	No data available	
Bulk Density	Not applicable	Liquid
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Glycerin	-1.75	
Citric acid	-1.72	
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
Viscosity	No data available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Other information

Section 10 - Stability and Reactivity

Reactivity

None known, based on information available

Stability Stable under recommended storage conditions.

Conditions to Avoid Heat, flames and sparks.

Incompatible Materials None known.

Hazardous Decomposition Products None under normal use conditions.

Hazardous Polymerization Hazardous polymerization does not occur.

Section 11 - Toxicological Information

Information on Toxicological Effects

Product Information Product does not present an acute toxicity hazard based on known or supplied information

(a) acute toxicity;

Oral

Based on ATE data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on ATE data, the classification criteria are not met

Based on available data, the classification criteria are not met

Based on ATE data, the classification criteria are not met

Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Glycerin	12600 mg/kg (Rat)	> 10 g/kg (Rabbit)	> 2.75 mg/L/4h (Rat)(mist)
D-Mannitol	LD50 = 13500 mg/kg (Rat)		
Tris (hydroxymethyl) aminomethane	LD50 = 5900 mg/kg (Rat)	LD50 > 5000 mg/kg (Rat)	
Sodium chloride	LD50 = 3 g/kg (Rat)	LD50 > 10000 mg/kg (Rabbit)	LC50 > 42 mg/L (Rat) 1 h
Polyoxyethylene(20)sorbitan monolaurate	LD50 = 37000 mg/kg (Rat)		LC50 > 5.1 mg/L (Rat) 4 h
Citric acid	LD50 = 3 g/kg (Rat)	>2 g/kg (Rat)	
[1,1'-Biphenyl]-4,4'-diamine, 3,3',5,5'-tetramethyl-	> 316 mg/kg (bird)		
Sulfuric acid	2140 mg/kg (Rat)		LC50 = 0.375 mg/L (Rat) 4 h
Sodium azide	LD50 = 27 mg/kg (Rat)	-	LC50 0.054 - 0.52 mg/L (Rat) 4 h
Ethylenediamine tetraacetic acid (EDTA)	4500 mg/kg (Rat) >2000 mg/kg (Rat)		1 mg/l (rat)
Urea peroxide	>2000 mg/kg (Rat)		
Potassium chloride	LD50 = 2600 mg/kg (Rat)		
Dihydrogen potassium phosphate	LD50 = 3200 mg/kg (Rat)	LD50 > 4640 mg/kg (Rabbit)	LC50 > 0.83 mg/L (Rat) 4 h

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	Australia	New Zealand	New South Wales	Western Australia	IARC	EU	UK	Germany
Sulfuric acid		Confirmed carcinogen			Group 1			

(g) reproductive toxicity; Category 1B
Developmental Effects May cause harm to the unborn child

(h) STOT-single exposure; Category 3
Results / Target organs Respiratory system

(i) STOT-repeated exposure; No data available
Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available

Section 12 - Ecological Information

Ecotoxicity effects

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Glycerin	LC50: 51 - 57 mL/L, 96h static (Oncorhynchus mykiss)			
Sodium chloride	Pimephals prome: LC50: 7650 mg/L/96h	EC50: 1000 mg/L/48h		
Citric acid	Leuciscus idus: LC50 = 440-760 mg/L/96h	EC50 = 120 mg/L/72h		Photobacterium phosphoreum: EC50 = 14 mg/L/15 min
Sulfuric acid	LC50: > 500 mg/L, 96h static (Brachydanio rerio)	EC50: 29 mg/L/24h	-	-
Sodium azide	LC50: = 0.7 mg/L, 96h (Lepomis macrochirus) LC50: = 0.8 mg/L, 96h (Oncorhynchus mykiss) LC50: = 5.46 mg/L, 96h flow-through (Pimephales promelas)			
Ethylenediamine tetraacetic acid (EDTA)	LC50: 34 - 62 mg/L, 96h static (Lepomis macrochirus) LC50: 44.2 - 76.5 mg/L, 96h static (Pimephales promelas)	EC50: = 113 mg/L, 48h Static (Daphnia magna)	EC50: = 1.01 mg/L, 72h (Desmodesmus subspicatus)	
Potassium chloride	Lepomis macrochirus: LC50: 1060 mg/L /96h Pimephales promelas:	EC50: 825 mg/L/48h	EC50: 2500 mg/L/72h	

	LC50: 750 - 1020 mg/L /96h			
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Persistence and Degradability**Persistence**

Soluble in water, Persistence is unlikely, based on information available.

Bioaccumulative Potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Glycerin	-1.75	No data available
Citric acid	-1.72	No data available

Mobility

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

Persistent Organic Pollutant

This product does not contain any known or suspected substance

Ozone Depletion Potential

This product does not contain any known or suspected substance

Section 13 - Disposal Considerations

Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

Contaminated Packaging

Dispose of in accordance with local regulations. Dispose of this container to hazardous or special waste collection point.

Other Information

Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer.

Section 14 - Transport Information

IMDG/IMO

UN-No	UN2796
Proper Shipping Name	Sulphuric acid
Technical Shipping Name	Mycoplasma hyopneumoniae ELISA Kit
Hazard Class	8
Packing Group	II

ADG

UN-No	UN2796
Proper Shipping Name	Sulphuric acid
Technical Shipping Name	Mycoplasma hyopneumoniae ELISA Kit
Hazard Class	8
Packing Group	II

Component	Hazchem Code
Sulfuric acid 7664-93-9 (STOP Solution)	2P 4WE 2W 2R
Sodium azide 26628-22-8 (0.1)	2XE
Urea peroxide 124-43-6 (0.05)	1X

IATA

UN-No	UN2796
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Proper Shipping Name	Sulphuric acid
Technical Shipping Name	Mycoplasma hyopneumoniae ELISA Kit
Hazard Class	8
Packing Group	II

Environmental hazards	No hazards identified
Special Precautions	No special precautions required
Additional information	None known

Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

National Regulations Australia

See section 8 for national exposure control parameters.

Standard for the Uniform Scheduling of Medicines and Poisons

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

Component	Standard for the Uniform Scheduling of Medicines and Poisons
Tris (hydroxymethyl) aminomethane - 77-86-1	Schedule 4 listed - in preparations for injection except in preparations containing ≤3% of Trometamol
Sulfuric acid - 7664-93-9	Schedule 6 listed - except its salts and derivatives; except in fire extinguishers, or in preparations containing ≤0.5% of Sulfuric acid
Ethylenediamine tetraacetic acid (EDTA) - 60-00-4	Schedule 4 listed - for human therapeutic use except in preparations containing ≤0.25% of Edetic acid; as Dicobalt edentate in preparations the treatment of Cyanide poisoning, or in contact lens preparations
Urea peroxide - 124-43-6	Schedule 5 listed - in preparations except in preparations containing ≤9% of Carbamide peroxide Schedule 6 listed - except: when included in Schedule 5, or in other preparations containing ≤9% of Carbamide peroxide Schedule 10 listed
Potassium chloride - 7447-40-7	Schedule 4 listed - in oral preparations for human therapeutic use except: a) when containing ≤550 mg of Potassium chloride per dosage unit, b) in preparations for oral rehydration therapy, c) in preparations for oral use for bowel cleansing prior to diagnostic medical and surgical procedures, or d) in preparations for enteral feeding
Dihydrogen potassium phosphate - 7778-77-0	Schedule 10 listed

Australian Industrial Chemicals Introduction Scheme (AICIS)

Component	Australian Industrial Chemicals Introduction Scheme (AICIS)	Additional information
Water - 7732-18-5	Present	-
Glycerin - 56-81-5	Present	-
D-Mannitol - 69-65-8	Present	-
Tris (hydroxymethyl) aminomethane - 77-86-1	Present	-
Albumins, blood serum - 9048-46-8	Present	-
Sodium chloride - 7647-14-5	Present	-
Polyoxyethylene(20)sorbitan monolaurate - 9005-64-5	Present	-
Citric acid - 77-92-9	Present	-
Phosphoric acid, disodium salt, dodecahydrate - 10039-32-4	Present	-
[1,1'-Biphenyl]-4,4'-diamine, 3,3',5,5'-tetramethyl- - 54827-17-7	Present	-
Sulfuric acid - 7664-93-9	Present	-
Sodium azide - 26628-22-8	Present	-

Ethylenediamine tetraacetic acid (EDTA) - 60-00-4	Present	-
Urea peroxide - 124-43-6	Present	-
Potassium chloride - 7447-40-7	Present	-
Dihydrogen potassium phosphate - 7778-77-0	Present	-

Australian - Illicit Drug Precursors/Reagents Substance List

This product contains one or more substance(s) on the Illicit Drug Precursors/Reagents list. Verify requirements related to using, handling and storing these substances.

Chemicals of Security Concern

This product contains one or more substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

Component	Australian - Illicit Drug Precursors/Reagents Substance List	Chemicals of Security Concern
Sulfuric acid - 7664-93-9	Category 3	Listed in Appendix A
Sodium azide - 26628-22-8		Listed in Appendix A Precursors to homemade explosives - concentration >=95%

Legend

Category 3 - Chemicals and apparatus that may be used in the illicit production of drugs. Purchases from this list should alert companies or organizations to seek further indicators of any suspicious orders or enquiries. No official reporting is required for items on this list unless considered warranted

Chemicals of Security Concern - for further information see <http://www.chemicalsecurity.gov.au/securityconcerns>

National pollutant inventory Subject to reporting requirements

Component	National pollutant inventory
Sulfuric acid - 7664-93-9	10 tonne/yr. Threshold category 1

Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

Component	Australia	New South Wales	Western Australia	New Zealand
Sulfuric acid - 7664-93-9				Confirmed carcinogen

International Inventories

Component	AICS	NZIoC	EINECS	ELINCS	TSCA	DSL	NDL	PICCS	ENCS	ISHL	IECSC	KECL
Water	X	X	231-791-2	-	X	X	-	X	X		X	KE-35400
Glycerin	X	X	200-289-5	-	X	X	-	X	X	X	X	KE-29297
D-Mannitol	X	X	200-711-8	-	X	X	-	X	X	X	X	KE-23061
Tris (hydroxymethyl) aminomethane	X	X	201-064-4	-	X	X	-	X	X	X	X	KE-01403
Albumins, blood serum	X	X	232-936-2	-	X	X	-	X	-		X	KE-05-0011
Sodium chloride	X	X	231-598-3	-	X	X	-	X	X	X	X	KE-31387
Polyoxyethylene(20)sorbitan monolaurate	X	X	-	-	X	X	-	X	X	X	X	KE-31681
Citric acid	X	X	201-069-1	-	X	X	-	X	X	X	X	KE-20831
Phosphoric acid, disodium salt, dodecahydrate	X	X	-	-	-	-	-	X	X		X	-
[1,1'-Biphenyl]-4,4'-diamine, 3,3',5,5'-tetramethyl-	X	X	259-364-6	-	X	X	-	-	-		X	-
Sulfuric acid	X	X	231-639-5	-	X	X	-	X	X	X	X	KE-32570
Sodium azide	X	X	247-852-1	-	X	X	-	X	X	X	X	KE-31357

Ethylenediamine tetraacetic acid (EDTA)	X	X	200-449-4	-	X	X	-	X	X	X	X	KE-13648
Urea peroxide	X	X	204-701-4	-	X	-	X	X	-		X	KE-35147
Potassium chloride	X	X	231-211-8	-	X	X	-	X	X	X	X	KE-29086
Dihydrogen potassium phosphate	X	X	231-913-4	-	X	X	-	X	X	X	X	KE-28622

Legend: X - Listed, '-' - Not Listed, R - Indicates a substance that is the subject of a Section 6 risk management rule under TSCA, XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)). **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

International Regulations

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Rotterdam Convention (PIC) Not applicable

Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

Component	Basel Convention (Hazardous Waste)	Australian Hazardous Waste Act - Categories of Wastes to Be Controlled
Citric acid - 77-92-9	Annex I - Y34	Y34 solid or solution
Sulfuric acid - 7664-93-9	Annex I - Y34	Y34 solid or solution

Component	CAS No	OECD HPV	Restriction of Hazardous Substances (RoHS)	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Water	7732-18-5	Listed	Not applicable	Not applicable	Not applicable
Glycerin	56-81-5	Listed	Not applicable	Not applicable	Not applicable
D-Mannitol	69-65-8	Not applicable	Not applicable	Not applicable	Not applicable
Tris (hydroxymethyl) aminomethane	77-86-1	Listed	Not applicable	Not applicable	Not applicable
Albumins, blood serum	9048-46-8	Not applicable	Not applicable	Not applicable	Not applicable
Sodium chloride	7647-14-5	Listed	Not applicable	Not applicable	Not applicable
Polyoxyethylene(20)sorbitan monolaurate	9005-64-5	Not applicable	Not applicable	Not applicable	Not applicable
Citric acid	77-92-9	Listed	Not applicable	Not applicable	Not applicable
Phosphoric acid, disodium salt, dodecahydrate	10039-32-4	Not applicable	Not applicable	Not applicable	Not applicable
[1,1'-Biphenyl]-4,4'-diamine, 3,3',5,5'-tetramethyl-	54827-17-7	Not applicable	Not applicable	Not applicable	Not applicable
Sulfuric acid	7664-93-9	Listed	Not applicable	Not applicable	Not applicable
Sodium azide	26628-22-8	Not applicable	Not applicable	Not applicable	Not applicable
Ethylenediamine tetraacetic acid (EDTA)	60-00-4	Listed	Not applicable	Not applicable	Not applicable
Urea peroxide	124-43-6	Not applicable	Not applicable	Not applicable	Not applicable
ProClin 300	NA	Not applicable	Not applicable	Not applicable	Not applicable
Potassium chloride	7447-40-7	Listed	Not applicable	Not applicable	Not applicable
Dihydrogen potassium phosphate	7778-77-0	Listed	Not applicable	Not applicable	Not applicable

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Citric acid	-	Use restricted. See item 75. (see link for restriction details)	-
Sulfuric acid	-	Use restricted. See item 75. (see link for restriction details)	-
Ethylenediamine tetraacetic acid (EDTA)	-	Use restricted. See item 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>

Section 16 - Other Information

Legend

AICS - Australian Inventory of Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

IECSC - Chinese Inventory of Existing Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

NZS 5433:2020 - Transport of Dangerous Goods on Land

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

WEL - Workplace Exposure Limit

DNEL - Derived No Effect Level

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

VOC - (Volatile Organic Compound)

NZIoC - New Zealand Inventory of Chemicals

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

CAS - Chemical Abstracts Service

ACGIH - American Conference of Governmental Industrial Hygienists
Predicted No Effect Concentration (PNEC)

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail

OECD - Organisation for Economic Co-operation and Development

LC50 - Lethal Concentration 50%

ATE - Acute Toxicity Estimate

RPE - Respiratory Protective Equipment

NOEC - No Observed Effect Concentration

BCF - Bioconcentration factor

PBT - Persistent, Bioaccumulative, Toxic

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Revision Date 14-Jul-2023

Revision Summary Update to GHS format.

This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).

Disclaimer

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End of Safety Data Sheet